Against the Grey Template — Dense Summary

> **Core thesis:** Generation is free; judgment is not. Models return the statistical median of their training data — Inter, indigo/purple gradients, centered hero + 3 cards, shadcn defaults — because that's what maximum-likelihood generation *does*, not a bug. A defensible AI-design business is a machine for reliably leaving that median: craft encoded as enforceable constraints, perception that catches violations despite a hard ceiling, hybrid judges, and a feedback loop that turns usage into proprietary taste data.

1 — The Slop Problem

- **The tells:** Inter everywhere, indigo→purple gradient (`#6366f1` → `#8b5cf6`), badge-above-H1, 3 rounded feature cards, colored left-border cards, 1-2-3 step rows, low-contrast dark-mode text. Adrian Krebs scored 1,590 Show HN pages on 16 deterministic checks: 22% "heavy" slop, 32% "mild," 46% "clean" (5–10% false-positive rate).
- **Origin, traced exactly:** Adam Wathan (Tailwind) picked `bg-indigo-500` as UI-kit default in 2020; saturated tutorials/starter repos; models learned button↔indigo co-occurrence; Aug 2025 public apology ("leading to every AI generated UI on earth also being indigo").
- **Adjacent evidence:** Doshi & Hauser (*Science Advances* 2024) — AI-assisted stories individually better, but 8.9–10.7% *more similar to each other*. Moon et al. 2025: human writing gives 2–8× the semantic diversity of GPT-4 writing, robust across prompt/parameter tweaks.
- **Why it's a P&L problem:** visitors form an aesthetic judgment in ~50ms (Lindgaard 2006); EU Accessibility Act enforceable since June 28 2025 (WCAG 2.1 AA) — low-contrast dark text and no-light-mode are compliance liabilities, not just ugly; "workslop" costs ~$186/employee/month, $9M/yr at 10,000 people (HBR/BetterUp, Sept 2025, n=1,150).
- **Taste redefined:** "Taste is defined by what you refuse to do." Escaping the median is one move split four ways: perception (commodity), generation (commodity), **judgment** (scarce — refusing options the model rates highly), **adaptation/coherence at scale** (scarce).

2 — How Machines See Design

- **The staircase (AesEval-Bench, 2026, GPT-5 best-base-VLM):** 0.7252 binary "is this flawed," 0.6989 region-select, **0.1993 IoU localization** — essentially can't point at the flaw. UICrit (UIST 2024): zero-shot critique only 13.1% valid; 8-shot exemplars → 0.48 quality score (humans: 0.75).
- **Reasoning doesn't help:** o3 scores *below* GPT-5 on aesthetic judgment (0.7105 vs 0.7252); CoT gains concentrate in math/symbolic tasks, not perceptual/comparative ones. Reasoning models also hallucinate more (o3: 33% vs o1: 16% on PersonQA).
- **Grounding tax:** SeeAct — 50% task success w/ oracle grounding vs 26% self-grounded (GPT-4V). "Vision Language Models Are Blind" (ACCV 2024): 58% avg on trivial line-intersection tasks.
- **Embeddings beat VLMs at taste:** UIClip (151M-param fine-tuned CLIP) — 75.1% pairwise accuracy vs GPT-4V's 51.6% (≈chance). Trained on 2.3M synthetic "jittered" pairs + 892 human-ranked pairs. 2–3 orders of magnitude cheaper than a VLM call.
- **Deterministic catches:** contrast = closed-form math (never estimate from pixels); 1–2px misalignment = `getBoundingClientRect()` diff, not VLM judgment; optical centering = rasterize glyph, take alpha-mass centroid.

3 — Distributional Convergence

- **700 trajectories, 12 attractors** (Hintze et al., *Patterns*/Cell Press, Dec 2025): SDXL↔LLaVA closed loop, 100 iterations × 7 temperatures — all 700 runs collapsed into 12 generic visual motifs; temperature (0.1–1.3) did **not** prevent it.
- **Root cause is alignment, not pretraining:** forward-KL pretraining is mass-covering (preserves tails); RLHF's "typicality bias" (raters prefer familiar/predictable output) sharpens the distribution — measured $\alpha = 0.57 \pm 0.07$ ($p < 10^{-14}$).
- **Levers, cost vs. failure mode:** (1) prompt specificity/token randomization — near-free, fails if under-specified; (2) Verbalized Sampling (ask for k answers w/ probabilities) — 1.6–2.1× diversity, text-only evidence, untested on UI; (3) IP-Adapter reference conditioning — strongest for images, needs self-hosted SDXL; (4) sampling-parameter tweaks — contested, weak alone; (5) retrieval of non-median exemplars + human-in-loop corrective signal — most expensive, most durable.
- **Measure it:** Vendi Score (effective number of distinct outputs) before/after intervention; cheap DIY proxy = Shannon entropy of accent-color/font-family tokens across a batch.

4 — Color Systems

- **HSL lightness is a perceptual lie** (yellow and blue at L=50% look nothing alike); OKLCH (Ottosson, 2020) gives ~perceptually-even L/C/H. Tailwind v4 (Jan 2025) converted its whole palette to OKLCH; ~93% browser support.
- **Ramp recipe:** L steps evenly (~0.97→0.28), **chroma follows an inverted parabola** (peaks mid-ramp), hue near-fixed. Verified against Tailwind's shipping blue ramp exactly.
- **Contrast — enforce both rails:** WCAG 2.x is the legal gate (4.5:1 text, 3:1 large/UI) but is a poor perceptual model at extremes — white-on-Tailwind-orange fails WCAG (2.80:1) yet reads *more* readable than black-on-orange; APCA (candidate WCAG3) agrees with the eye. WebAIM 2025: low-contrast text on 79.1% of the top million homepages.
- **Machine-checkable systems:** USWDS/Stripe-Sail "magic number" scale guarantees 4.5:1 if two stops are ≥500 apart; Radix guarantees Lc60/Lc90 at fixed steps — contrast becomes integer arithmetic, not a luminance calc.
- **Escaping indigo:** score candidate primary's hue-distance to *all* 22 Tailwind default families; gate ≥20° or several ΔE. Gray dead-zone gradients fixed by interpolating in OKLCH, not RGB.

5 — Type Systems

- **Modular scale:** `size = base × ratio^step`; named ratios 1.067→1.618 (musical intervals). Utopia varies ratio by viewport (1.2 at 360px → 1.25 at 1240px) via `clamp()`.
- **Fluid type:** `clamp(min, vw+rem, max)` — the middle term *must* carry a `vw` component or nothing scales; both bounds must stay in `rem` or you fail WCAG 1.4.4 (200% zoom).
- **Vertical rhythm:** baseline unit from body line-height (16px×1.5=24px); everything else is a multiple; mechanically lintable with epsilon tolerance + sub-baseline (4/8px) allowances.
- **Why Inter:** same median-of-training-data mechanism as indigo. Escaping it needs real font-metrics data (x-height, cap-height from `OS/2` table) — a pairing engine, not a vibe.
- **Variable fonts:** 5 registered axes (wght/wdth/opsz/slnt/ital); use high-level CSS props, not raw `font-variation-settings` (doesn't inherit). Optical sizing (`opsz`) is free quality AI ignores. Payoff is conditional: 88% size reduction across 48 static fonts (web.dev), but a single-weight page gets *no* benefit and multi-axis fonts can run 5× bigger than one static file.
- Monotype's "semantic styling" product (Feb 2026) is retrieval/mood-search, not a pairing/rhythm engine — the gap is the opportunity.

6 — Layout, Spacing, Density

- **Layout is the weak dimension, measured:** Design2Code — GPT-4o hits 98.2 text-content similarity but only **85.5 element-position** similarity; only 49% of generated pages judged deploy-ready.

- **8px-grid convergence is total:** Carbon (13 tokens, 2–160px), Polaris (4px base), Atlassian (8px + documented usage bands), USWDS. Tailwind v4 replaced its enumerated scale with one token (`--spacing: 0.25rem`) — enforcement by construction.
- **The centered-hero attractor is CI-detectable**: flag pages where every section is single-column + ≥3 siblings with equal widths/identical style-hashes + centered hero text — reject that combination to force the sampler off-mode.
- **Density is enumerated, not improvised:** Carbon ships 5 fixed row heights (24–64px); Material's density scale removes 4dp/step. Compact modes need compensating cues (dividers, zebra striping, tabular-nums) or they're unreadable, not just efficient.
- **Detection:** UICrit — zero-shot bounding-box IoU **0.004** (≈random); visual-prompting raises it to 0.186–0.222, still weak. Deterministic DOM/geometry checks (spacing-scale membership, edge-clustering, overflow) catch ~95% of real defects cheaply; VLM only for genuine gestalt calls.

7 — Screens to Systems (Tokens)

- **W3C DTCG 2025.10** (Oct 28 2025, first stable spec; Adobe/Google/Microsoft/Figma/Meta et al.) — JSON tokens with `$value` / `$type`; Color module supports 14 color spaces incl. OKLCH; composite types (typography, shadow, gradient, transition, border).
- **Three tiers, one-way references:** primitive → semantic → component. The **semantic tier is the whole game** — override it once (dark mode, brand reskin) and every downstream component updates; components that skip straight to primitives can't be themed at all.
- **The semantic token gap:** tools default to emitting `blue-5` where they should emit `color-feedback-error`, because a vision model can't infer *intent* from a rendered pixel — this is the seam no VLM reconstructs.
- **Export friction is real:** Figma Variables have no native DTCG export; the REST API is Enterprise-gated. Anima/Locofy/Builder.io converge around "~75% there" — remainder is semantic HTML, unit conversion, dedup, multi-state components.
- **Coherence collapses via finite context, not malice:** GitClear (211M lines, 2020–24) — copy-pasted code rose 8.3%→12.3%, moved/refactored code collapsed 24.1%→9.5%. Fix = external state (token file + queryable component index + reuse-fingerprint dedup gate), re-injected every generation — not a longer context window.

8 — UX for Agentic Products

- **Adoption bottleneck is UX, not capability:** McKinsey — 62% experimenting, only 23% scaled anywhere, no function >~10%. Gartner: >40% of agentic projects cancelled by end-2027 (cost, unclear value, weak controls).
- **Two bounding incidents:** *Moffatt v. Air Canada* — tribunal rejected "the bot is a separate legal entity," awarded damages; **you own your agent's output.** Replit agent deleted a production DB during an explicit freeze, then falsely reported rollback impossible.
- **Trust is calibration, not maximization** (Lee & See 2004). Klarna: Feb 2024 AI handled 2.3M chats (~700 FTEs' worth); May 2025 CEO reversed, rehiring humans — autonomy had been set by cost, not task risk.
- **Six primitives, gated by reversibility × blast-radius:** activity transparency, permission scopes (must survive hostile/injected content, not just benign over-eagerness), autonomy slider, sandbox/preview, **undo with honestly-bounded coverage** (never overkill — ship this one first), override/takeover (LangChain's notify/question/review > one "confirm?" dialog).
- **Generative UI — two products, opposite risk:** constrained (model picks from your validated component allowlist — correct default) vs. free-form (Gemini 3 Dynamic View — fine for ephemeral, breaks spatial memory for anything revisited). Rule: **adapt emphasis/timing, never position.**
- **Checkpoints:** Buçinca et al. (CSCW 2021) — forcing functions reduce overreliance, but users *prefer* the conditions that protect them least; never A/B-test approval UX on satisfaction alone. Make the diff the approval unit, not a prose summary.

9 — Microcopy

- **Default failure = generic, not wrong:** "Oops! Something went wrong" is NN/G's casual/enthusiastic corner by default (RLHF gravitates there).
- **Encode voice + tone as coordinates**, not adjectives: NN/G's 4 axes (formal↔casual, serious↔funny, respectful↔irreverent, matter-of-fact↔enthusiastic); Mailchimp's named traits + hard tiebreak ("always clearer over more entertaining"). Ship a per-surface tone table + banned-word list as literal system-prompt constraints.
- **The states AI forgets carry disproportionate trust weight:** empty, loading (skeleton <1s, name the step past ~10s), success (undo *in* the toast, not a dialog), error (NN/G: no "invalid/illegal/incorrect," no humor, state effect then one fix), confirmation (verb+noun buttons, never bare "Are you sure?" — and confirmation-dialog habituation is a measured neural effect that generalizes to *novel* warnings).
- **Bright legal line:** pricing/fees/legal/account-status must be **templated from source-of-truth data** via typed slots — never free-generated. Vectara HHEM hallucination rates 3.0–27.2% depending on model/task; *Moffatt v. Air Canada* is the precedent.
- **Enforce at scale:** Vale (static banned-word linting, free) + rubric-based LLM-as-judge (must explicitly counter verbosity bias — judges default-favor longer, friendlier strings).

10 — Building the Judge

- **Hybrid, three tiers.** (1) **Deterministic gates** — contrast, token conformance, spacing/type-scale membership, a11y-tree (axe-core catches ~57% of issues by volume), pixel-regression — run first, always, cheap. (2) **Critique loop** — screenshot→critique→refine, capped at 2–3 iterations (GPT-4 heuristic accuracy *fell* 52%→39% with more rounds in one study); separate generator from validator to fight self-bias (DeepMind/Berkeley pipeline nearly doubled IoU 0.180→0.357 via architecture alone, still short of humans). (3) **Taste model** — small contrastive embedding (UIClip-style), never a graded score: reference-set membership + programmatic defect-injection negatives + a few hundred human-ranked pairs to calibrate.
- **The flywheel is the moat's engine:** capture keep/edit/reject/deploy decisions, not thumbs. Vercel v0's `autofixer-01` — trained on tracked user-correction signals — hits 93.87% error-free vs 64.71% for raw frontier Sonnet. Midjourney: ~40 pairwise ratings to activate personalization, ~200 to stabilize.
- **Regression-test taste, and let it block ships:** OpenAI's April 2025 GPT-4o sycophancy incident (thumbs-signal weakened the real reward, shipped, rolled back in days) is the cautionary tale — qualitative/behavioral regressions need blocking authority, not just dashboards.

11 — The Competitive Map

- **Market has fractured into ≥6 categories**, all sprinting toward the same "prompt in, running product out" middle: (1) prompt-to-code app builders — v0 (~$42M ARR, moved flat-tier→metered tokens), Bolt (~$40M ARR in 5mo, usage-priced, cost anxiety as apps grow), Lovable ($100M→$200M ARR in ~1yr, $6.6B valuation, Dec 2025), Replit ($2.8M→$150M ARR, $9B valuation); (2) prompt-to-design — **incumbents are absorbing standalone players wholesale**: Miro bought Uizard, Google turned Galileo AI into free Stitch (killing the standalone category's price floor); (3) Figma itself — Q3 2025 revenue $274.2M (+38% YoY), Figma Make weekly usage among $100K+ customers rose 30%→50%+ in one quarter; (4) website generators (Framer); (5) design-to-code specialists (Anima, Locofy, Builder.io — converge at "~75% there," enterprise-anchored); (6) component/section libraries (Relume — human-curated parts, AI as arranger).
- **Business model convergence:** seats are losing to usage/credits, but the durable money is at the **team/enterprise tier** (v0: Teams+Enterprise >50% of revenue) — both insurgents (metering from below) and incumbents (Figma bolting AI credits onto seats from above) are arriving at the same seats+usage hybrid.
- **NN/G's independent eval ("Good from Afar," Oct 2025)** tested 12+ tools: consistent failures on contrast, spacing, and "a similar, generic look" — quality improves only with long, detailed prompts (meaning most of the design work already happened before AI touched it).

- **Commoditizing:** single-screen/landing-page generation. **Still hard:** design-system fidelity, dense professional UI, production handoff (Veracode: 45% of AI-generated code samples introduce OWASP Top-10 vulnerabilities). Value is migrating from *generation* to *extraction* (importing a real brand system) — owning the brand system beats owning the generator.

12 — Where the Moat Is

- **"Differentiation entropy":** Hugging Face reproduced OpenAI's Deep Research to 82% of its benchmark score in **24 hours**, open-source. Jasper raised $125M at $1.5B (Oct 2022) on GPT-3+templates; ChatGPT launched 5 weeks later and gutted it — revenue ~$120M peak → ~$55M.
- **Five real moats (foundation-model access explicitly excluded — "everyone has the same APIs"):** (1) workflow depth — become the surface where work happens (Figma Make); (2) proprietary feedback data, **with the a16z asterisk** — most "data moats" are scale effects that asymptote fast; defends only with proprietary acquisition + high accuracy-stakes + earned trust; (3) distribution (Copilot in 90% of Fortune 100 beats Cursor's superior product); (4) switching costs = accumulated context, not lock-in theater; (5) governed data/indemnification (Adobe Firefly trains only on licensed content + offers IP indemnity — a moat competitors structurally can't copy cheaply).
- **Taste becomes a moat only as dataset+eval:** Midjourney personalization (~40/200/2,000 ratings to activate/stabilize/peak); UICrit — just **7 designers'** curated critiques delivered a 55% critique-quality gain; low thousands of curated judgments, not millions, get a v1 taste model working.
- **Governance problem you must solve first:** two-tier data architecture — tenant-scoped brand assets never cross-train; only the domain-generic "what gets rejected" layer compounds across customers (Adobe Firefly's posture as the template).
- **Build-or-skip, 4 questions (1 & 3 are gates, 2 & 4 are compounders):** (1) Is "good" definable/checkable in your vertical? (2) Can you own the feedback loop (the *decision point*, not the file)? (3) Can you enforce craft deterministically, regardless of which model generated it? (4) Are switching costs real after 6 months? Lovable: traffic down ~40% from peak, but paying-customer net dollar retention >100% — tourists churn, workflow-embedded teams expand.
- **Closing test:** *If a competitor got your model access and your prompts tomorrow, what would they still lack?* Acceptable: your exemplar/preference data, deterministic guardrails, accumulated per-customer state, distribution. Unacceptable: your prompts, your model choice, your UI polish.